

Row 2: Cash Flow Analysis & Financial Planning (Chapter 4)

Scenario

"GreenTech Innovations" is a high-growth startup that reports a net income of \$2.5 million for 2024 (accrual basis). However, its cash balance has been declining. The following data are available:

- Depreciation and amortisation: \$800,000
- Increase in accounts receivable: \$1.6 million
- Increase in inventories: \$2.0 million
- Increase in accounts payable: \$500,000
- Capitalised R&D expenditure (paid in cash): \$1.2 million (amortised over 5 years)
- Gain on sale of investments (non-cash): \$900,000
- Purchase of fixed assets: \$3.0 million
- Repayment of long-term debt: \$600,000
- Dividends paid: \$300,000

The firm's sales are expected to grow 30% next year, but because of supply chain bottlenecks, the CFO forecasts that inventory will need to be stocked 60 days earlier than usual, requiring an additional cash outlay of \$700,000 upfront in January. Sales are seasonal: 40% in Q1, 20% in Q2, 20% in Q3, and 20% in Q4. All sales are on credit; 60% is collected in the quarter following the sale, 40% in the quarter after that.

Questions

(a) Prepare GreenTech's Statement of Cash Flows for 2024, clearly showing the net cash provided by (used in) operating, investing, and financing activities. Explain why the firm's net income does not translate into a healthy cash balance.

(b) Prepare a monthly cash budget for Q1 of 2025, taking into account the accelerated inventory purchase, the sales forecast (Q4 2024 sales were 5million),andthecollectionpattern.Assumenoootherchangesandthatthefirmstarts withacashbalanceof5million),andthecollectionpattern.Assumenoootherchanges andthatthefirmstartswithacashbalanceof1.0 million. What is the minimum cash balance during Q1, and when does a potential shortfall occur?

(c) Discuss the limitations of the cash budget in (b) and outline how the firm could improve its cash management, specifically addressing working capital practices and the financing of inventory buildup.